



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2/ IP YEAR 6 2017**

CANDIDATE
NAME

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CENTRE
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INDEX
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CLASS

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H2 BIOLOGY

Paper 2 Structured Questions

9744/02

**Tuesday 12 September 2017
2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, Centre number, index number and class in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graph.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Q1	/10
Q2	/10
Q3	/12
Q4	/11
Q5	/12
Q6	/12
Q7	/11
Q8	/12
Q9	/10
Total	/100

This document consists of **24** printed pages and **2** blank pages.

Paper 2 (Part I)
Answer **all** the questions in this section.

- 1** Fig. 1.1 shows an electron micrograph of a plant cell.

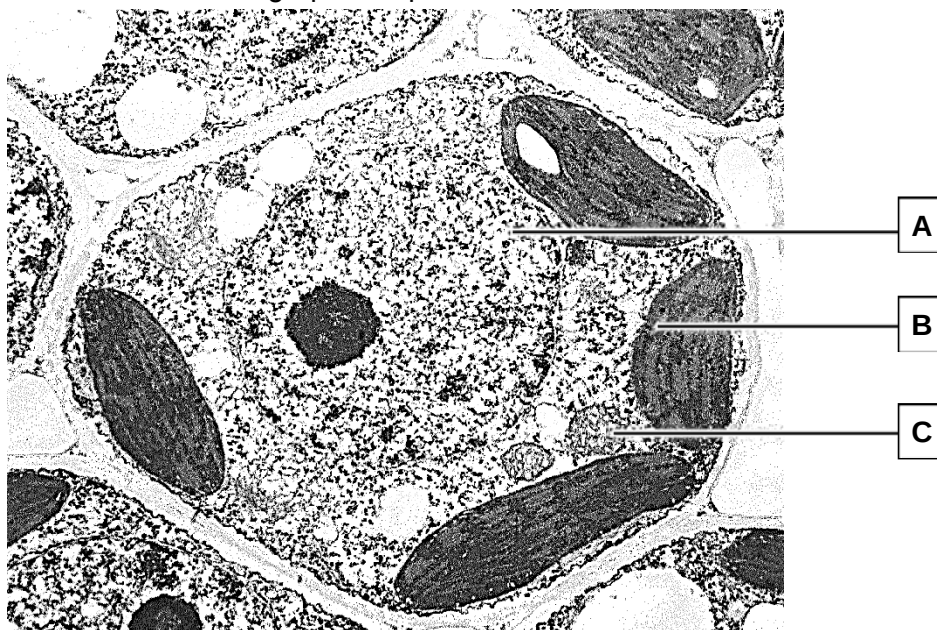


Fig. 1.1

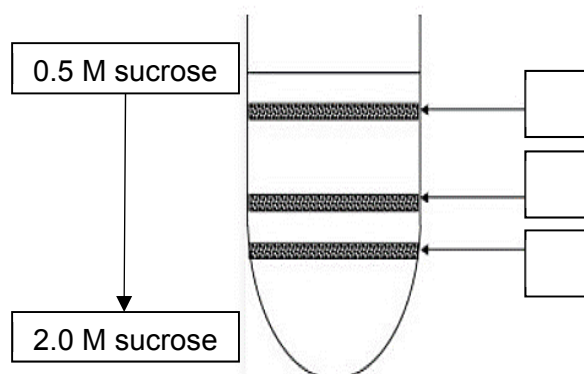
- (a)** Identify organelles **B** and **C**.

Organelle **B**: _____ [1]

Organelle **C**: _____ [1]

- (b)** Extracts from the homogenised plant cells in Fig. 1.1 were added to a sucrose density gradient and centrifuged at high speed to separate the various organelles.

- (i)** Label the bands where organelles **A**, **B** and **C** can be found after centrifugation.



[3]

(ii) Explain your answer in (b)(i).

[2]

In a separate experiment, protoplasts (plant cells with cell wall removed) were first treated with three different reagents – ethanol, distilled water and buffer solution, for two hours. The treated cells were then subjected to the density gradient centrifugation.

Fig. 1.2 shows the thickness of the lowest band for each type of treated cell after density gradient centrifugation.

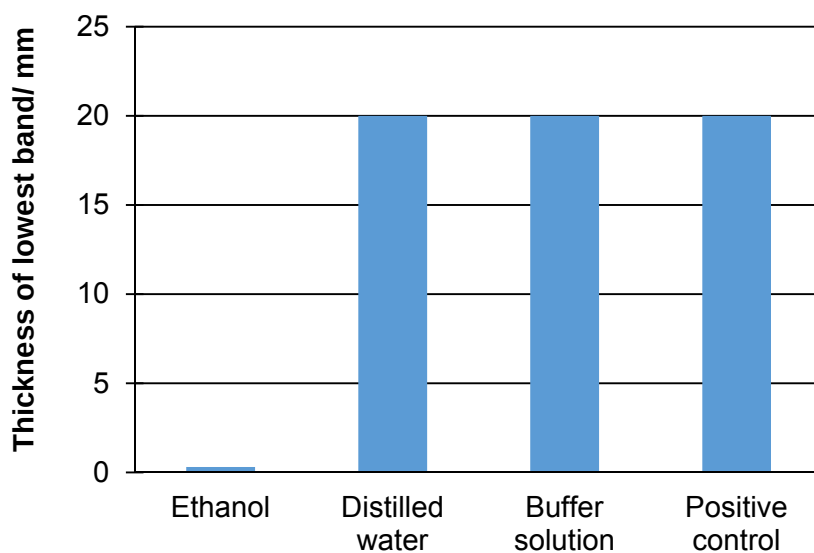


Fig. 1.2

(c) Explain the effects of the different reagents on the thickness of the lowest band.

[3]

[Total: 10]

- 2 The membrane composition of bacterial species varies according to the environmental conditions to which the cells are exposed to, thus allowing them to thrive in a wide range of environment. Bacterial membranes consist of a large diversity of *amphiphilic* lipids, including sphingolipids (general structure is shown in Fig. 2.1).

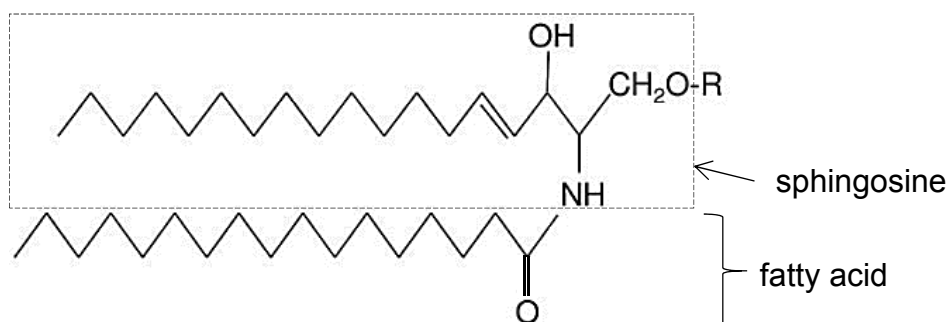


Fig. 2.1

- (a) Define the term *amphiphilic*.

[1]

- (b) (i) State **two** differences between the structure of sphingolipids and phospholipids.

[2]

- (ii) Draw a labelled diagram of the molecular structure of a phospholipid.

[3]

- (iii) Explain how the structure of phospholipid makes it a suitable component of cell membranes.

 [2]

Fig. 2.2 shows another component found in animal cell membranes.

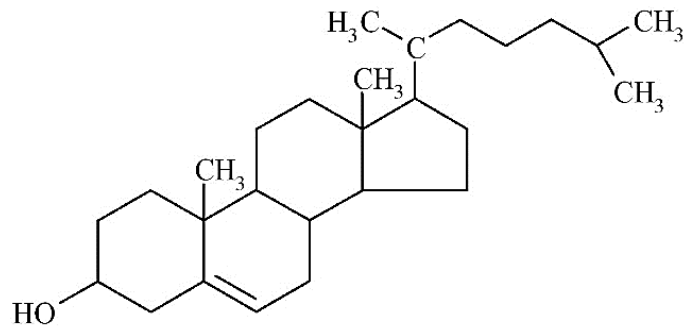


Fig. 2.2

- (c) Explain how the molecule shown in Fig. 2.2 performs its function in cell membranes.

 [2]

[Total: 10]

- 3 Fig. 3.1 shows a photomicrograph (Magnification x 5000) of two human cells, **A** and **B**, at different stages of the mitotic cell cycle.

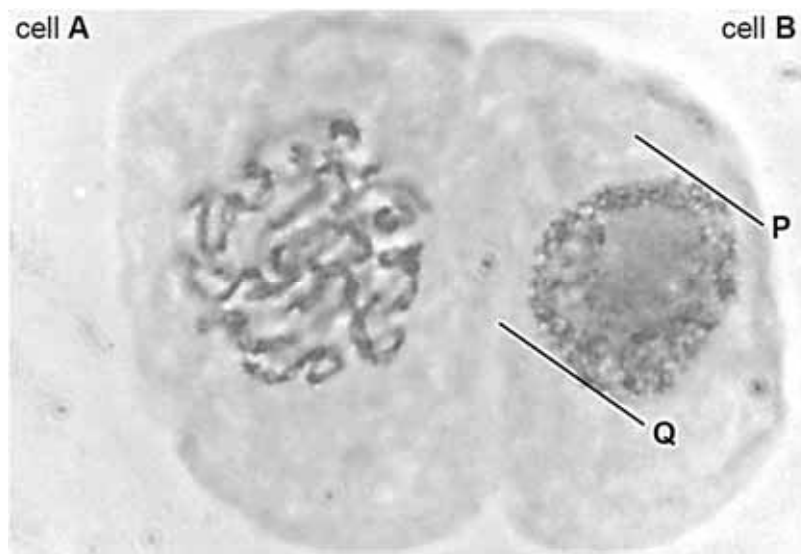


Fig. 3.1

- (a) (i) Name the stage of mitosis that is occurring in cell **A**.

_____ [1]

- (ii) Describe the events that are occurring in cell **A**.

 _____ [3]

- (b) Determine the actual length between lines **P** and **Q** in cell **B**.

[2]

Prostate cancer is one of the common cancers in males. It has been found that loss-of-function mutations to the DNA mismatch repair genes, such as *MSH2*, is associated with advanced prostate cancer. *MSH2* is a protein that dimerizes with another protein, *MSH6*, to form the DNA repair enzyme complex. *MSH2* has three domains – protein-binding domain, DNA-binding domain and ATPase domain (a region that removes phosphate group from ATP).

- (c) Explain how loss-of-function mutations in *MSH2* can lead to cancer development in the prostate.

[3]

Hormone therapy is typically used to treat prostate cancer but this treatment is only effective for a few years, after which resistance to the hormone will develop.

A study was carried out to determine the effectiveness of a chemotherapy drug, melphalan, in treating men with hormone-resistant prostate cancer. The side effect of melphalan is that formation of blood cells will be inhibited.

- (d) Suggest how the patient's own bone marrow can be used to counter the side effect of melphalan.

[3]

[Total: 12]

- 4 (a) Explain why mRNA is formed as a continuous strand during transcription while one of the DNA strands is formed discontinuously during replication.

[3]

Several types of rRNA and tRNA are transcribed as a single strand precursor RNA. Each rRNA (16S, 23S, 5S) and tRNA molecule is cleaved following transcription in a process known as RNA trimming (Fig. 4.1) to form mature rRNA and tRNA molecules.

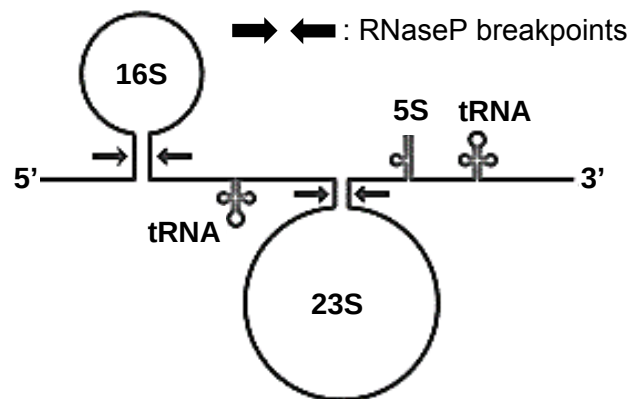


Fig. 4.1

- (b) State where rRNA genes are found.

[1]

- (c) Compare between the processes of RNA trimming and post-transcriptional modification for mRNA.

[3]

- (d) Relate how the single-stranded structure of rRNA and tRNA facilitates their roles.

[4]

[Total: 11]

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For Examiner's Use	
Q5	/12
Q6	/12
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H2 BIOLOGY

9744/02

Paper 2 (Part II)

- 5 Operons in bacteria allow them to regulate their gene expression in response to changes in the environmental conditions.

In order to investigate the function of the regulatory and structural genes of *lac* operon, loss-of-function mutation was induced in the sequences of various genes. The different effects of the mutation on the expression of *lac* genes are shown in Table 5.1.

Table 5.1

Region of DNA sequence in which gene mutation occurs	Allolactose absent		Allolactose present	
	β -galactosidase	transacetylase	β -galactosidase	transacetylase
A	+	+	+	+
B	–	–	–	–
C	–	–	–	+
D	–	–	+	–

(+) indicates the synthesis of functional enzyme

(–) indicates no synthesis of functional enzyme

- (a) (i) Identify regions **A** and **D**.

A: _____

D: _____ [2]

- (ii) Outline the effect of the mutation of region **A** on the expression of *lac* genes.

[2]

Mammals respond to changes in the environmental conditions using different mechanisms. For instance, blood glucose concentration can be regulated by hormones such as insulin and glucagon.

Fig. 5.1 shows the modification of preproinsulin to form insulin in organelles **X** and **Y**.

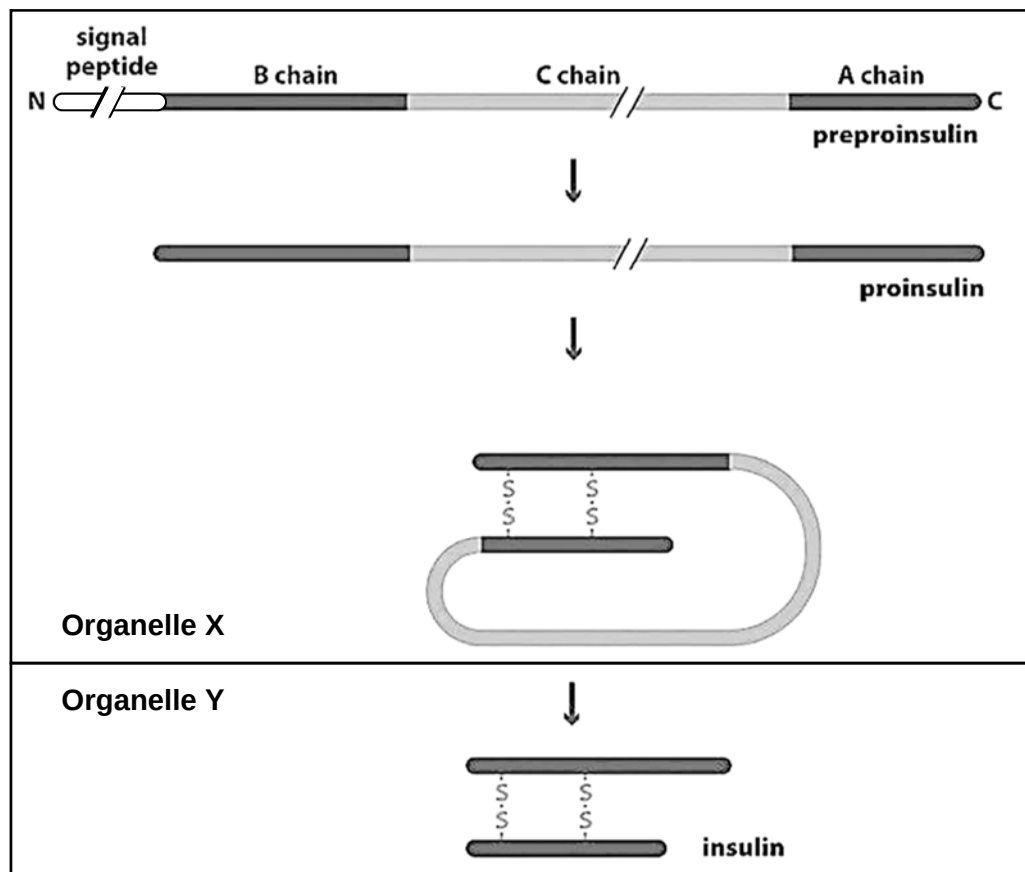


Fig. 5.1

- (b) With reference to Fig. 5.1, outline what happens in organelles **X** and **Y**.

[2]

Fig. 5.2 shows the effect of glucose on a pancreatic cell.

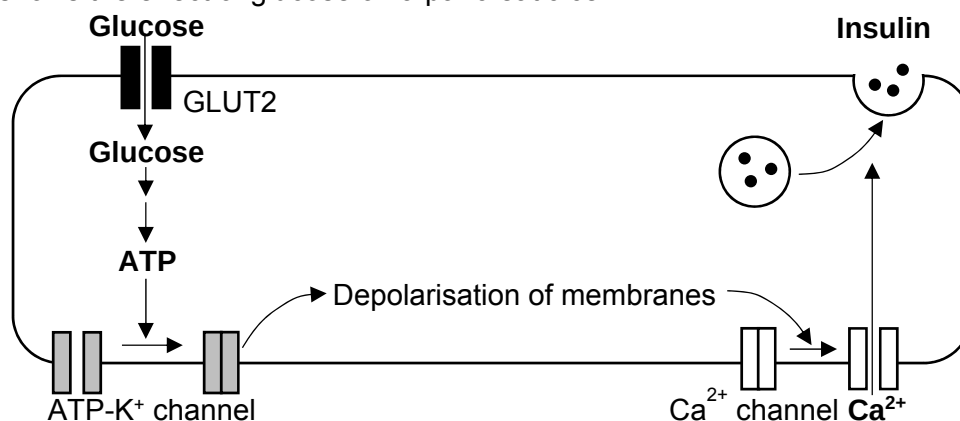


Fig. 5.2

- (c) With reference to Fig. 5.2, outline how the pancreatic cell responds to elevated blood glucose levels.

[3]

Mammalian hormones can be synthesized artificially using bacterial cells.

- (d) Suggest **one** problem associated with expressing mammalian genes in bacterial cells.

[1]

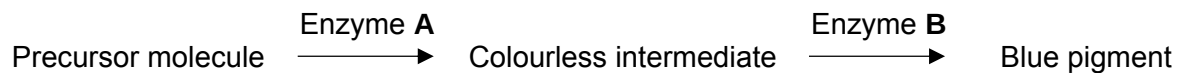
- (e) Compare the advantages of a mammalian response to changes in blood glucose concentration with that of a bacterial response to changes in supply of lactose.

[2]

[Total: 12]

- 6 Harebell flowers can be found in various habitats in Scotland and Finland.

The colour of harebell depends on the anthocyanin pathway in which colourless intermediates are converted to a blue pigment. Two genes coding for two different enzymes, **A** and **B**, are crucial in this biochemical pathway. Each gene has two alleles, with the dominant alleles coding for functional enzymes while the recessive alleles coding for non-functional enzymes.



Two different white-petal homozygous lines of harebells were crossed and all the F_1 plants had blue flowers.

- (a) Using symbols **A** and **B**, draw a genetic diagram to explain the cross above and the result of selfing F_1 .

The experimental results for the F_1 cross is shown in Table 6.1.

Table 6.1

	Blue flowers	White flowers
Observed numbers	12	9

A statistical test was carried out and a calculated value of 2.17 was obtained.

χ^2 – Distribution Table

Degrees of freedom	Probability, <i>p</i>				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

***t*-table**

Degrees of freedom	Probability, <i>p</i>				
	0.1	0.05	0.025	0.01	0.005
One-tailed t-test					
Two-tailed t-test	0.2	0.10	0.050	0.02	0.010
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.08	2.518	2.831

(b) State a suitable conclusion for the experiment.

[3]

(c) Explain the genetic basis for the observed results.

[3]

[Total: 12]

- 7 Glucose and fructose are two common fruit sugars used in winemaking. Another sugar used in the fermentation industry is sucrose. The effects of the three sugars on fermentation by yeast were investigated and the results are shown in Fig. 7.1 and Fig. 7.2.

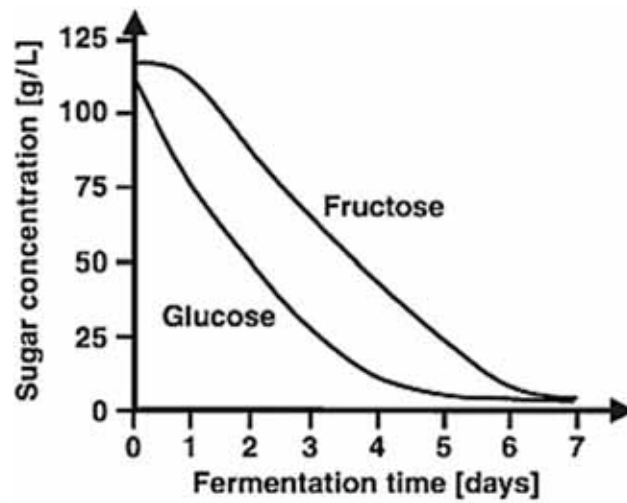


Fig. 7.1

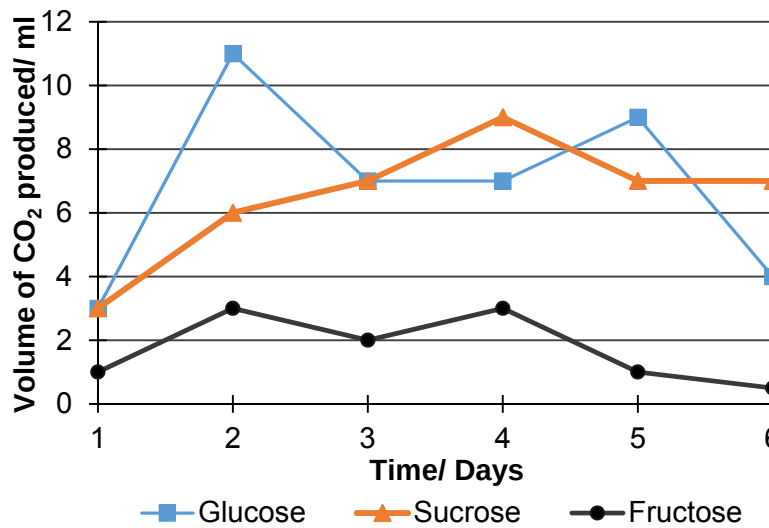


Fig. 7.2

- (a) Describe how ethanol is formed by yeast.

[2]

- (b) With reference to Fig. 7.1 and Fig. 7.2, explain the order in which the sugars were utilized by yeast for fermentation.

[4]

Fig. 7.3 shows a respirometer.

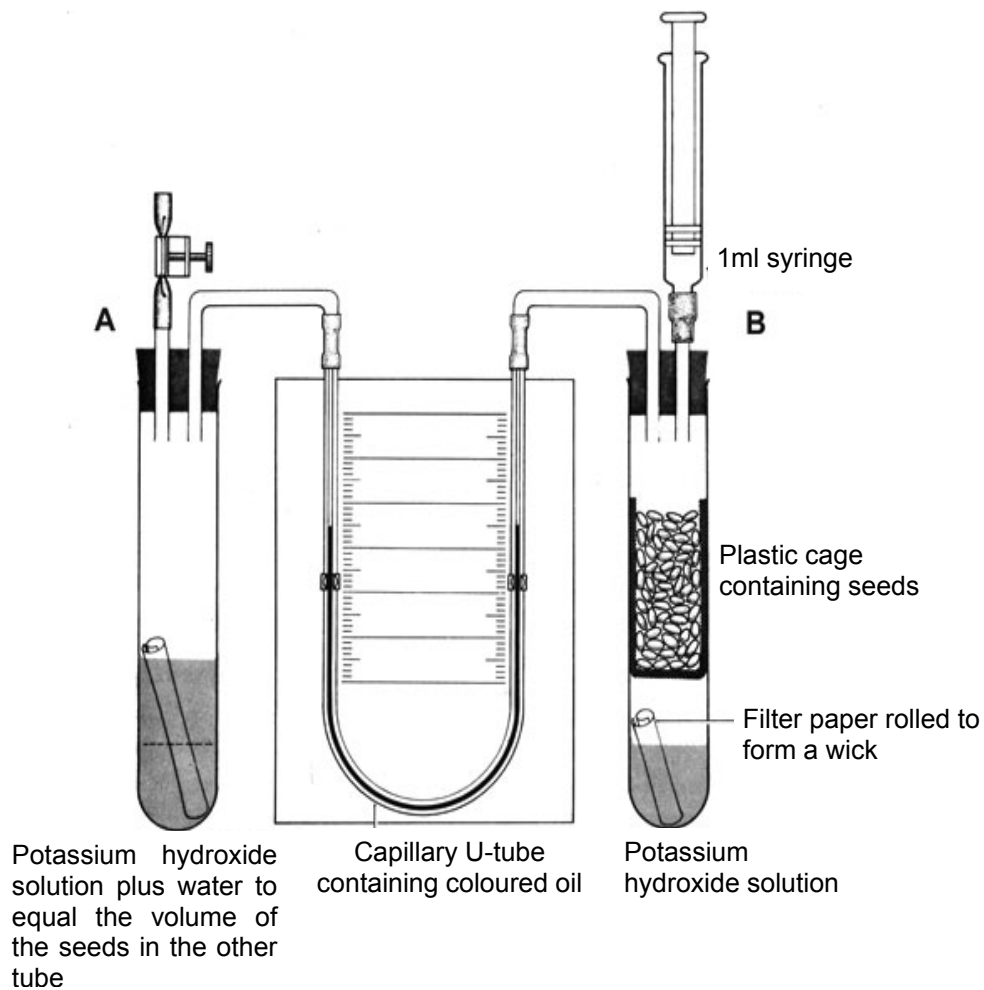


Fig. 7.3

- (c) Briefly explain how you can determine the rate of respiration using the set-up shown in Fig. 7.3.

[2]

- (d) Suggest how the compensation point of a plant will be affected when it undergoes anaerobic respiration.

[3]

[Total: 11]

- 8 Human Immunodeficiency Virus (HIV) emerged as a mysterious new disease in the early 1980s.

Fig. 8.1 shows the structure of HIV.

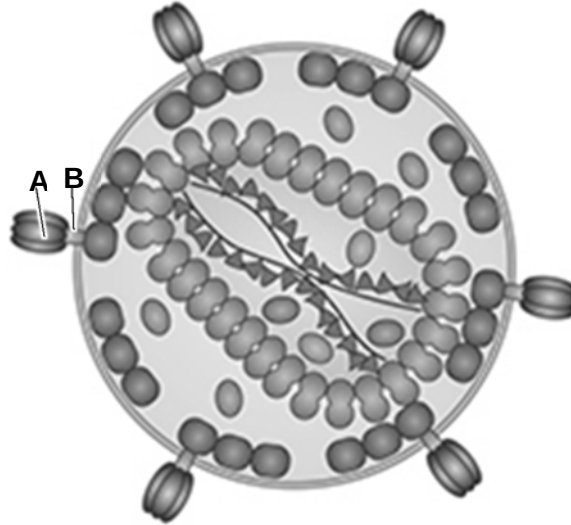


Fig. 8.1

- (a) Identify and describe the function of structures **A** and **B**.

[3]

In July 1990, a young woman in Florida, who had no known risk factors for HIV infection and no known contact with other HIV-positive persons, was tested HIV positive after undergoing an invasive dental procedure performed by a dentist who had Acquired Immunodeficiency Syndrome (AIDS).

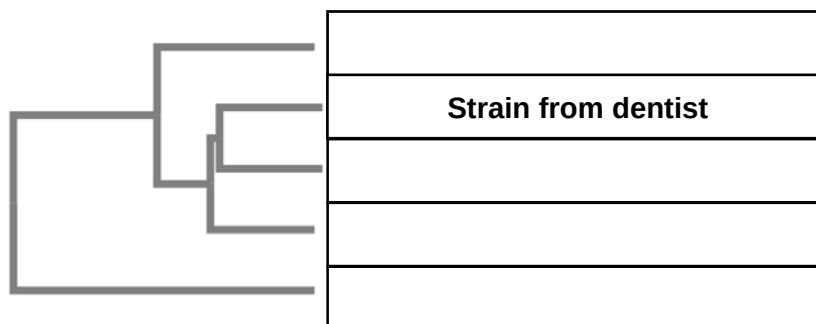
The U.S. Centre for Disease Control and Prevention (CDC) carried out an epidemiological investigation using DNA isolated from white blood cells of the dentist and patients **A** to **D**. The DNA was sequenced and compared with the strain isolated from the dentist.

Fig. 8.2 shows the multiple sequence alignment before a phylogenetic analysis could be carried out.

Strain from	Sequence																			
Dentist	-	-	-	-	C	-	T	A	-	T	T	G	-	C	T	G	G	C	G	C
Patient A	-	-	G	-	C	-	C	A	-	T	A	G	-	C	T	A	G	C	G	C
Patient B	-	-	G	-	C	A	C	C	-	T	-	G	-	C	T	A	G	C	G	C
Patient C	-	-	G	-	C	-	T	-	-	T	G	G	G	C	T	G	G	C	G	C
Patient D	C	A	G	A	C	-	T	A	C	T	-	G	-	C	T	A	G	-	G	-

Fig. 8.2

(b) Complete the figure below to show how closely related the different strains of HIV are.



[2]

In order to determine the origins of HIV, researchers conducted a similar phylogenetic analysis. Molecular epidemiology data showed that Simian Immunodeficiency Virus (SIV), which infects 36 species of primates found in sub-Saharan Africa, crossed over to infect human population known to eat wild animals. There are two types of HIV: HIV-1 and HIV-2.

Fig. 8.3 shows the phylogenetic tree of HIV / SIV.

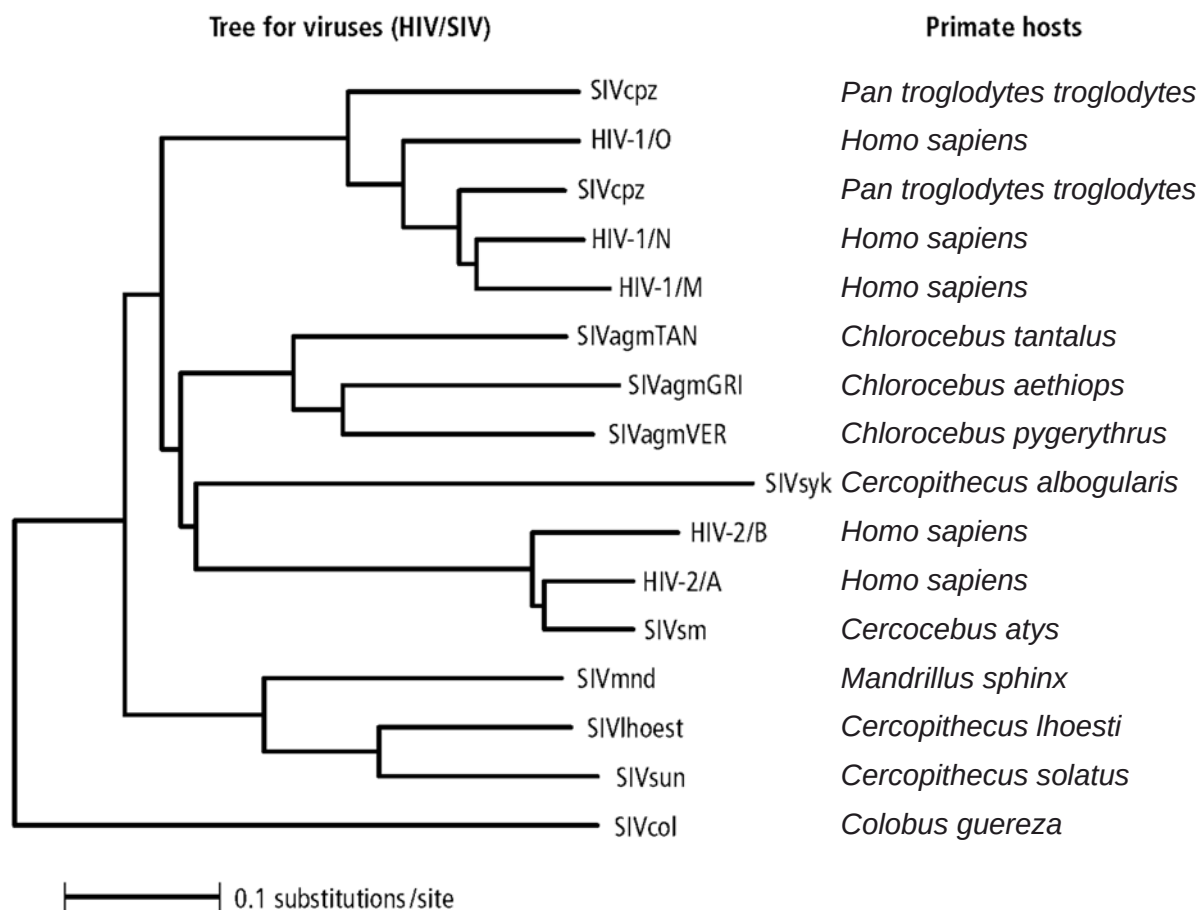


Fig. 8.3

(c) (i) Describe how HIV-1 and HIV-2 are evolutionarily related.

_____ [1]

(ii) Identify the closest viral strains and their primate hosts of HIV-1 and HIV-2.

_____ [2]

SIV may have been around for more than a million years in these primates, but they do not cause disease in the primates. However, when SIV infected humans, it evolved to form HIV which causes AIDS in humans.

- (d) (i)** Explain how HIV arose from SIV, allowing it to cause disease in humans.

[3]

- (ii)** Suggest the selective advantage for not causing death in the primate host.

[1]

[Total: 12]

- 9 Tetanus is an infection caused by a Gram-positive bacterium, *Clostridium tetani*. The bacteria can exist in the dormant state as endospores. The endospores can be found everywhere in the environment, including soil, dust, and manure. They can enter the body through broken skin, usually through injuries from contaminated objects, and resume their active state.

(a) (i) Describe the structure of the cell wall in *C. tetani*.

[2]

(ii) Suggest the advantage of forming of endospores.

[1]

(b) Outline how the innate immune system responds to infection by *C. tetani*.

[3]

When *C. tetani* invades the body, a toxin is released into bloodstream. The toxin causes painful muscular contractions. It can also cause breathing problems, severe muscle spasms, seizures, and paralysis.

Tetanus vaccine was developed as a prophylactic measure against *C. tetani* toxins. The vaccine is produced by subjecting the bacterial toxins to chemical treatment to inactivate the toxins.

(c) Explain how tetanus vaccine can prevent the clinical symptoms.

[4]

[Total: 10]

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